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MINI PROJECT REPORT
ON

“Smart vaccum cleaner”

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CERTIFICATE

Certified that the mini project work entitled Smart vacuum cleaner carried out by Ms. Rishika Singh, /Ms. Akhila Nippuleti, Mr./Ms. Tanisha Santosh, USN: 1CR24EC172, USN: 1CR24EC133, USN: 1CR24EC218, bonafide students of CMRIT in partial fulfillment For the award of Bachelor of Engineering / Bachelor of Technology in ELECTRONICS AND COMMUNICATION of the Visveswaraiah Technological University, Belgaum during the year 2025-2026 It is certified that all corrections/suggestion indicated for Internal Assessment have been incorporated in the Report. The mini project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

Signature of the Guide

Signature of the HOD

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ABSTRACT

This project presents the design and development of a Smart Vacuum Cleaner Robot using Arduino Uno and low-cost recyclable materials. The robot is capable of automatic movement and obstacle detection using an ultrasonic sensor. A vacuum cleaning mechanism powered by a DC motor helps in collecting dust and small particles from the floor surface. The system uses Arduino-based control connected with motor drivers, gear motors, a servo motor, and sensors to navigate efficiently while avoiding collisions. Going beyond conventional cleaning devices, this proposed system integrates obstacle avoidance with an innovative autonomous waste disposal unit to minimise manual household human intervention.

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Chapter 1

INTRODUCTION

1.1 Introduction

Automation and robotics are becoming increasingly important in modern households. Smart cleaning robots reduce manual effort and improve convenience. This project focuses on designing a compact smart vacuum cleaner robot using Arduino Uno.

1.2 Motivation and Background

Maintaining cleanliness is essential but can be time-consuming. Commercial robotic vacuum cleaners are often expensive. This project aims to create a low-cost alternative using easily available electronic components and waste materials.

1.3 Objectives

- To design a smart vacuum cleaner robot using Arduino Uno.
- To implement automatic obstacle detection using an ultrasonic sensor.
- To develop a simple vacuum cleaning mechanism.
- To reduce human effort in floor cleaning.

Chapter 2

LITERATURE REVIEW

The field of autonomous floor cleaning robotics has seen significant progression, transitioning from purely random path execution to intelligent, sensor-driven systems. Researchers have explored various control paradigms, navigation algorithms, and mechanical designs to maximize dust collection efficiency while minimizing cost and power overhead. This section evaluates recent methodologies implemented in microcontroller-based cleaning systems.

An autonomous vacuum cleaning prototype developed by K. Sarath Kumar et al. [a] focuses on spatial evaluation using pre-programmed decision logic on an Arduino platform. When the vehicle encounters an obstruction, its sensor loop determines the direction with optimal boundary clearance and commands the drivetrain to turn toward that open space. The physical design relies on a lithium battery configuration driving a high-RPM axial fan coupled directly to a motor. To measure physical operational parameters, the authors incorporated an anemometer and a pressure gauge, concluding that a detachable collection bag improves overall suction efficiency. Meanwhile, long-distance communication methods were introduced by Humayun Rashid et al. [b] through the implementation of a Dual-Tone Multi-Frequency (DTMF) control architecture. This mechanism allows a cell phone to manipulate the room cleaner from unrestricted distances over cellular frequencies. The robot operates via two distinct smart path-following methods to ensure complete area coverage, and features an upgraded vacuum system equipped with specific filtration units to manage airflow safely.

To bridge the gap between localized control and automation, K.A.M. Annuar et al. [c] integrated an HC-06 Bluetooth module with an Arduino controller, creating an interface that links seamlessly with an Android smartphone. This structure offers dual operational flexibility, letting the user toggle between a self-directed autonomous mode and a manual mode where movement is directed in real time via a custom mobile application. In contrast to manual orientation, P. Pravin et al. [d] targeted algorithmic path efficiency by introducing a border analysis technique. Instead of traversing a room arbitrarily, this high-pressure automatic vacuum robot maps out and traverses the boundaries of an area first, subsequently adopting a systematic zigzag trajectory to cover the remaining floor space. The system analyzes obstacles using an array of ultrasonic sensors and cleans using a specialized high-pressure suction pump. Further refining the hybrid control scheme, D. Desmulyati et al. [e] designed an Arduino R3-based autonomous vacuum cleaner capable of dynamic obstacle avoidance. Their system pairs localized smartphone control via Bluetooth with a physical, chassis-mounted analog ON/OFF push button, providing an immediate manual override option for safety and operational testing.

Chapter 3

PROBLEM STATEMENT AND METHODOLOGY

3.1 Problem Statement

Manual floor cleaning requires effort and time. Commercial robotic vacuum cleaners are expensive. This project develops a low-cost smart vacuum cleaner robot capable of automatic navigation and cleaning.

3.2 Proposed Methodology

Phase 1:

Structural Fabrication & Assembly The robot chassis base is constructed utilising strong, lightweight, recyclable plastic lids and waste materials. The vacuum collection chamber is sculpted using a shaped plastic bottle acting as a focused venturi tube. A secondary servo motor is linked to a trapdoor flap at the back of the chamber to manage the autonomous waste unloading sequences.

Phase 2:

Circuit Integration & Calibration The Arduino Uno is stacked with the L293D Motor Shield, creating localised high-current nodes to supply power to the four independent DC gear motors. The sensory array is wired up and filtered against high-frequency electromagnetic noise caused by the high-RPM suction core fan.

Phase 3:

Navigation & Decision Logic An Embedded C algorithm is deployed to maintain continuous distance evaluation pings at 10ms rates. When an obstacle drops within a critical 20cm threshold, the controller cuts drive power to the wheels, activates the panning servo to map clearance metrics on the Left (45°) and Right (135°) quadrants, and selects the path with the largest unblocked threshold.

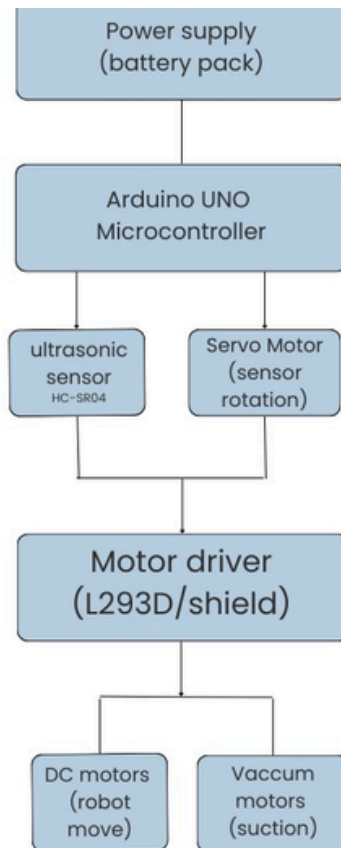
Phase 4:

Integration Testing & Verification The full system undergoes cleaning efficiency runs with debris markers (fine dust, paper trimmings). The autonomous dumping hatch cycle is validated to check structural containment and clean mechanical open/close motions.

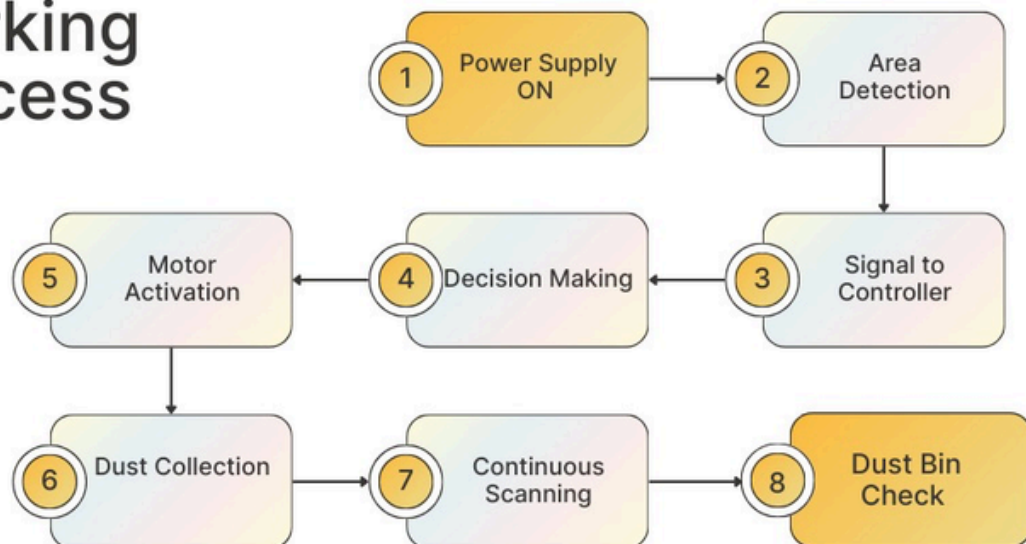
3.3 Working Principle

The ultrasonic sensor continuously measures the distance between the robot and nearby objects. If an obstacle is detected, the Arduino processes the data and changes the movement direction of the robot automatically.

Block diagram



Working process



Chapter 4

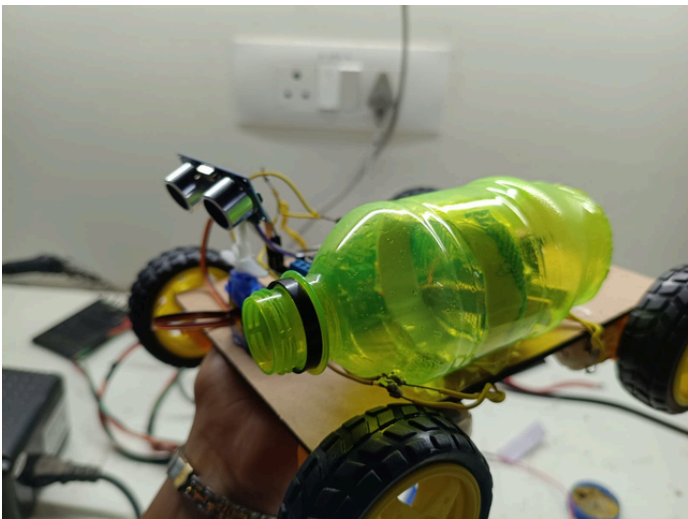
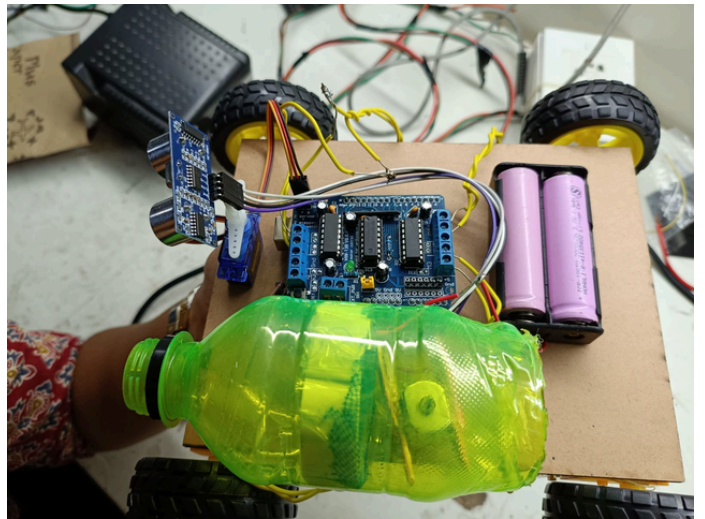
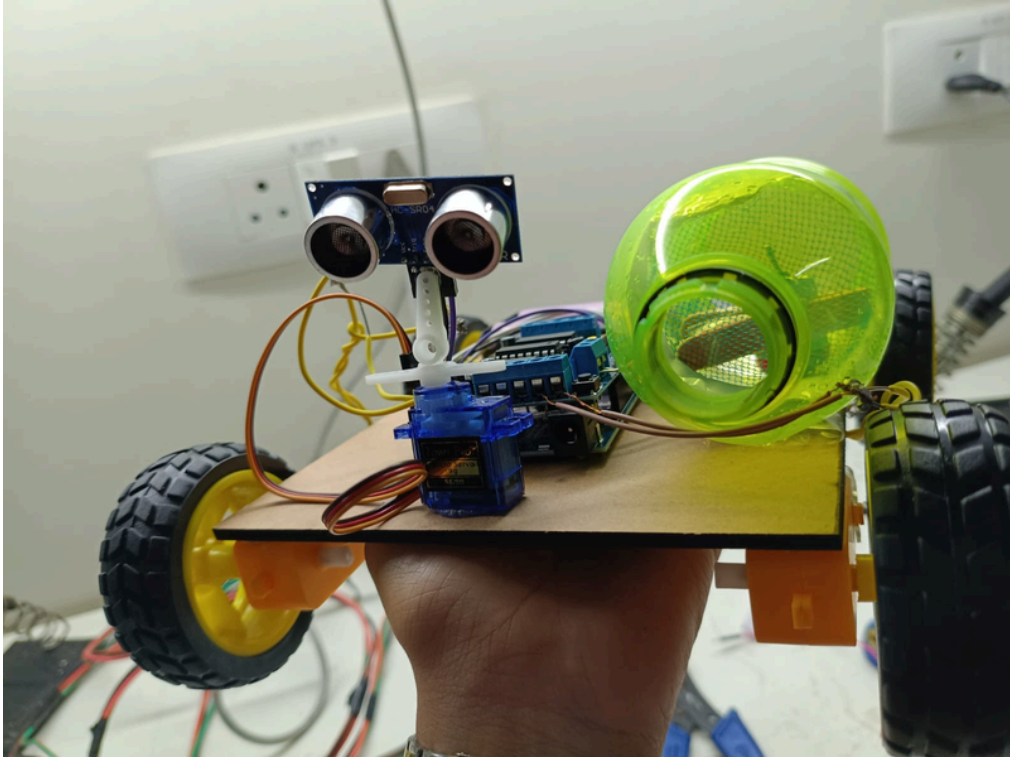
SYSTEM IMPLEMENTATION

4.1 Hardware Requirements

- **Arduino Uno R3:** Microcontroller core for running algorithmic logic.
- **L293D Motor Driver Shield:** Controls wheel speed and rotational direction vectors.
- **Four DC Gear Motors & Wheels:** Provides consistent indoor locomotion torque.
- **HC-SR04 Ultrasonic Sensor:** Provides obstacle distance measurement data.
- **SG90 Servo Motor:** Provides 180° sweeps for spatial obstacle mapping.
- **12V High-RPM DC Vacuum Motor:** Generates localised negative pressure for dust capture.
- **Lithium-ion Battery Pack:** High-discharge power cell network.
- **Chassis Materials:** Plastic bottle container, plastic structural lids, intake pipe inlet.

4.2 Software Requirements

- **Arduino IDE:** Local development environment for compilation.
- **Embedded C Programming Language:** Written script architecture managing peripheral loops.



Chapter 5

IMPLEMENTATION AND RESULTS

5.1 Implementation

The robot chassis was created using waste plastic materials. Gear motors were attached for movement and connected to the motor driver shield. The ultrasonic sensor and servo motor were mounted at the front for obstacle scanning.

5.2 Results

The smart vacuum cleaner robot successfully moved across the floor while avoiding obstacles. The vacuum motor collected lightweight dust particles effectively.

Chapter 6

APPLICATIONS AND ADVANTAGES

6.1 Applications

- Residential indoor floor maintenance and dry-sweeping.
- Low-cost educational robotics demonstrations and engineering mini-projects.
- Foundational prototyping for smart home appliance networks.

6.2 Advantages

- Low-Cost Implementation: Uses readily available hobby components and recyclable plastic materials rather than proprietary industrial parts.
- Eco-Friendly Fabrication: Structural integration of recycled plastic housings reduces overall e-waste footprint.
- True Autonomy: The inclusion of self-disposal mechanical structures transforms the platform from a semi-autonomous appliance into a hands-free utility tool.

Chapter 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The Smart Vacuum Cleaner Robot using Arduino was successfully designed and implemented. The prototype achieves efficient autonomous navigation, obstacle detection, and active suction containment. By incorporating a dedicated waste-release system, the platform effectively minimises human maintenance overhead, proving that budget-friendly consumer hardware can solve common domestic household chores.

7.2 Future Scope

- Implementing Wi-Fi or Bluetooth modules to support remote mobile data telemetry.
- Upgrading from basic ultrasonic sweeps to advanced LiDAR or SLAM systems for precise indoor floor mapping.
- Integrating an automated charging dock with a matching waste-collection bin to support autonomous battery replenishment and automated debris transfer.

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